

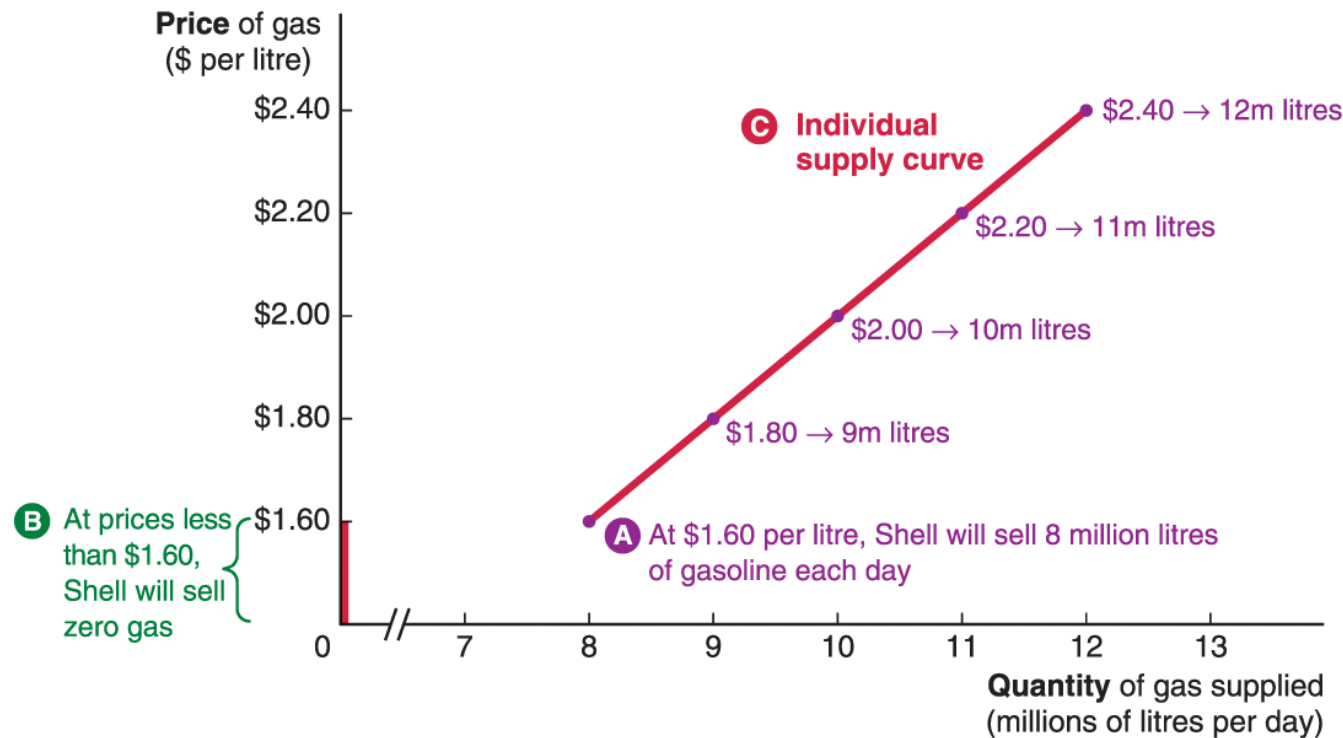
Chapter 3: Supply and Producer Choice

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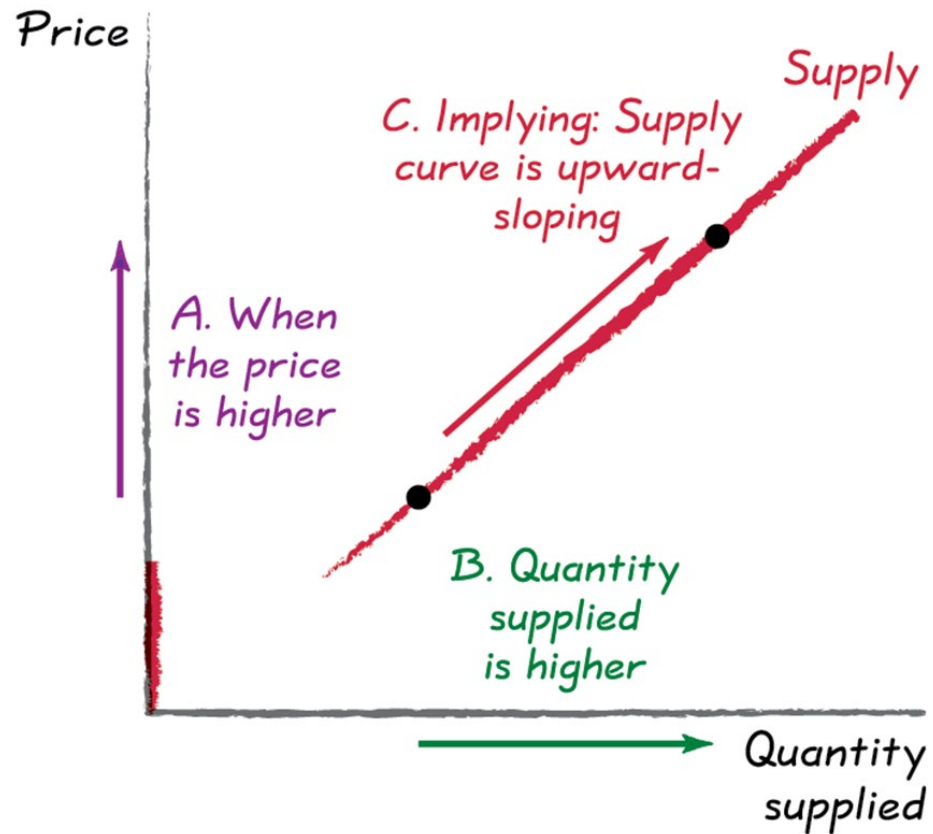
UNIVERSITY OF WATERLOO

Shell's individual supply curve



- A** When the price is **\$1.60 per litre**, Shell supplies **8 million litres** per day to the market. If the price rises to **\$1.80 per litre**, the quantity supplied increases to **9 million litres** per day.
- B** At price below \$1.60 per litre, Shell will stop supplying gas.
- C** The **individual supply curve** shows the quantity supplied by Shell at each price.

Law of supply

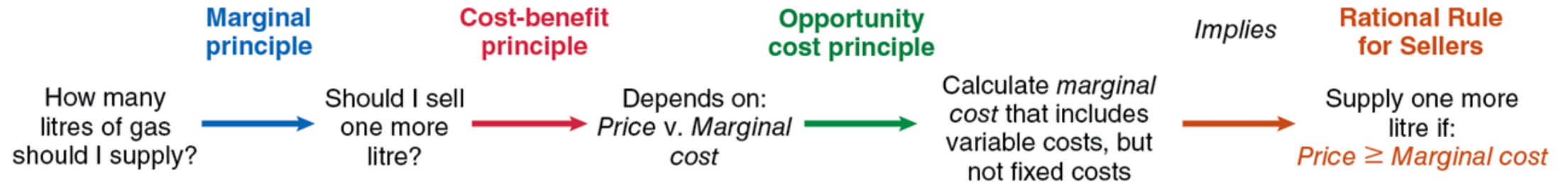


- The **law of supply** says that the quantity supplied of a good decreases when its price decreases and increases when its price increases.
- In other words, individual supply curves are upward sloping.
- The supply curve shows what happens to supply when the price changes **holding everything else constant** (“ceteris paribus”).

Perfect competition

- In Chapter 2, we assumed Darren and all the other buyers in the gasoline market do not choose the price. In this chapter, we similarly assume that gas suppliers are **price takers** (not price makers). How do we justify this assumption?
- A market is **perfectly competitive** if:
 1. many buyers and sellers (each small relative to the market) and no barriers to entry and exit
 2. all firms sell an identical product
 3. perfect information
 4. zero mobility costs (costs nothing for buyers to switch from one seller to another)
- In a perfectly competitive market, there's no incentive for firms to sell at a price that deviates from their competitors' price. Why? Because if they sell at a higher price, nobody will buy from them, and if it was possible to sell at a lower price and still make profit, a competitor would already have exploited that opportunity.
- In the real world, perfectly competitive are rare, but it simplifies the theoretical analysis.

Choosing the best quantity to supply



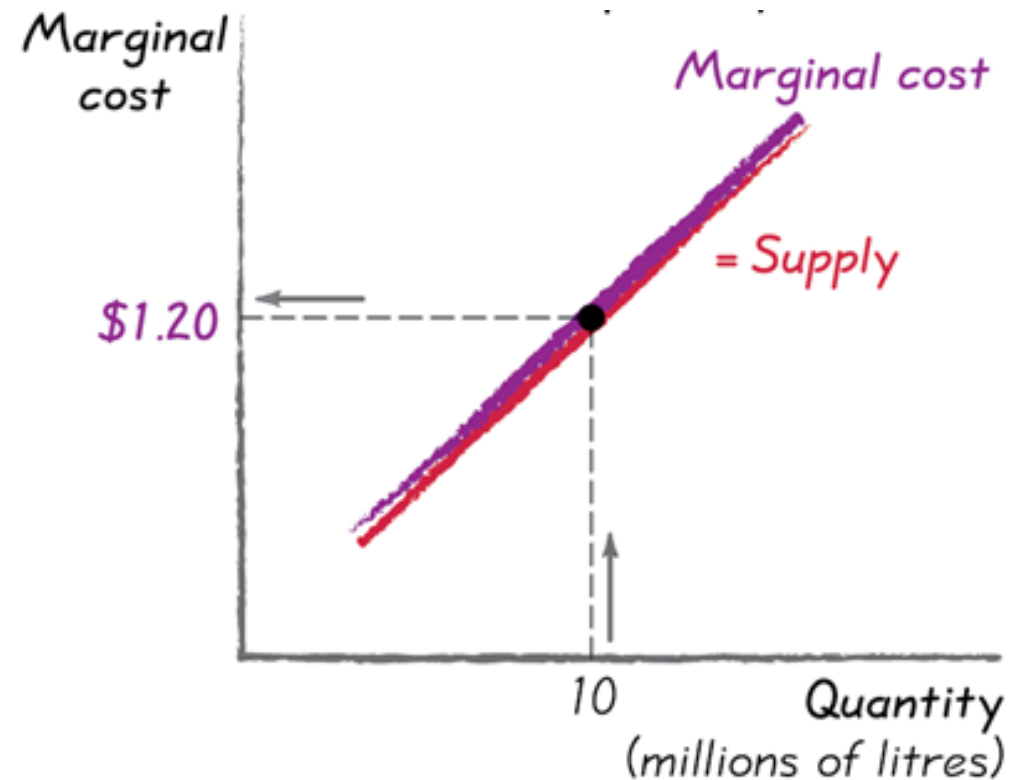
- The **marginal benefit** of selling one more litre of gas is the market price.
 - To calculate the **marginal cost** of selling one more unit, Shell needs to apply the opportunity cost principle – how do costs compare when the extra litre is not produced?
 - The marginal cost of production includes **variable costs** which vary with the quantity of output produced (e.g., labour), but not **fixed costs**, which do not vary with the quantity of output (e.g., capital).
- ➡ **Sell one more unit if the price is greater than (or equal to) the marginal cost.**

The rational rule for sellers

- The rational rule says to keep selling more until the **marginal benefit** equals the **marginal cost**.
- Since the marginal benefit is the price of a litre of gas, Shell should sell until the marginal cost is equal to the **price**.

$$\text{Price} = \text{Marginal cost}$$

- The law of supply is a consequence of **diminishing marginal product** and **rising input costs**.



Diminishing marginal product

- **Marginal cost** tends to increase as more output is produced. Why?
- As a firm uses more of a variable input (e.g., labour) with a given quantity of fixed inputs (e.g., kitchen equipment), the **marginal product of the variable input** diminishes.
- Diminishing marginal product means that marginal increases in output become increasingly difficult to achieve and so the **marginal cost of production** increases.



Mathematical example

$$Y = \sqrt{L \times K} \quad (\text{"production function"})$$

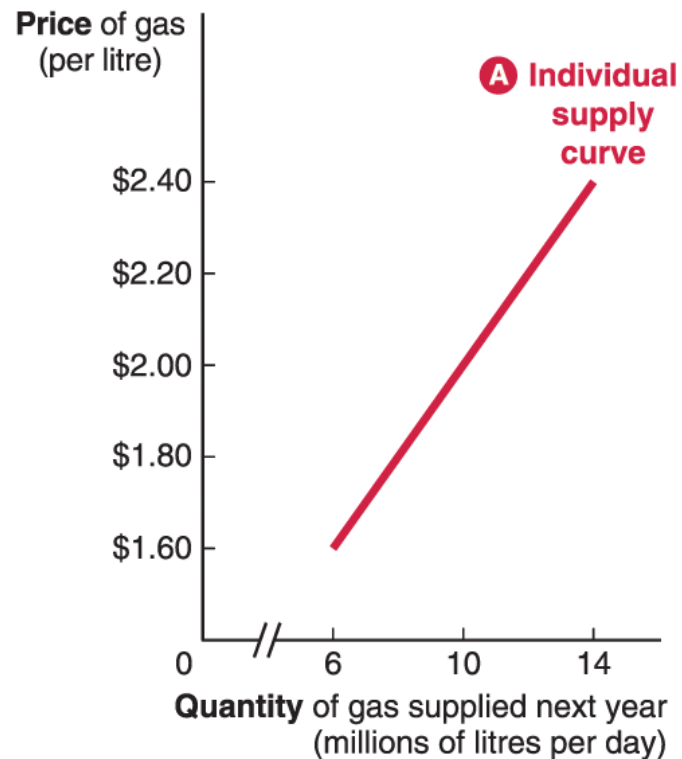
L ("labour")	K ("capital")	Y ("output")	MP_L
0	4	0	--
1	4	2	2
2	4	2.83	0.83
3	4	3.46	0.64
4	4	4.00	0.54
5	4	4.47	0.47
6	4	4.90	0.43

Mathematical example

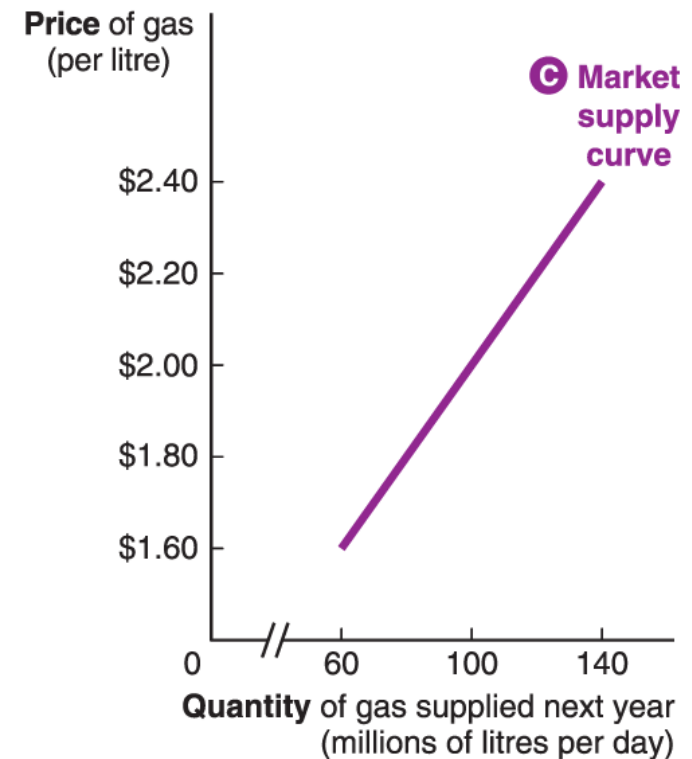
$$\text{If } Y = \sqrt{L \times K}, \text{ then } L = \frac{Y^2}{K}$$

<i>Y</i>	<i>K</i>	<i>L</i>	<i>TC</i> (<i>wage</i> =\$15)	<i>MC</i>
0	4	0	0	--
1	4	0.25	3.75	3.75
2	4	1	15	11.25
3	4	2.25	33.75	18.75
4	4	4	60	26.25
5	4	6.25	93.75	33.75
6	4	9	135	41.25

From individual to market supply

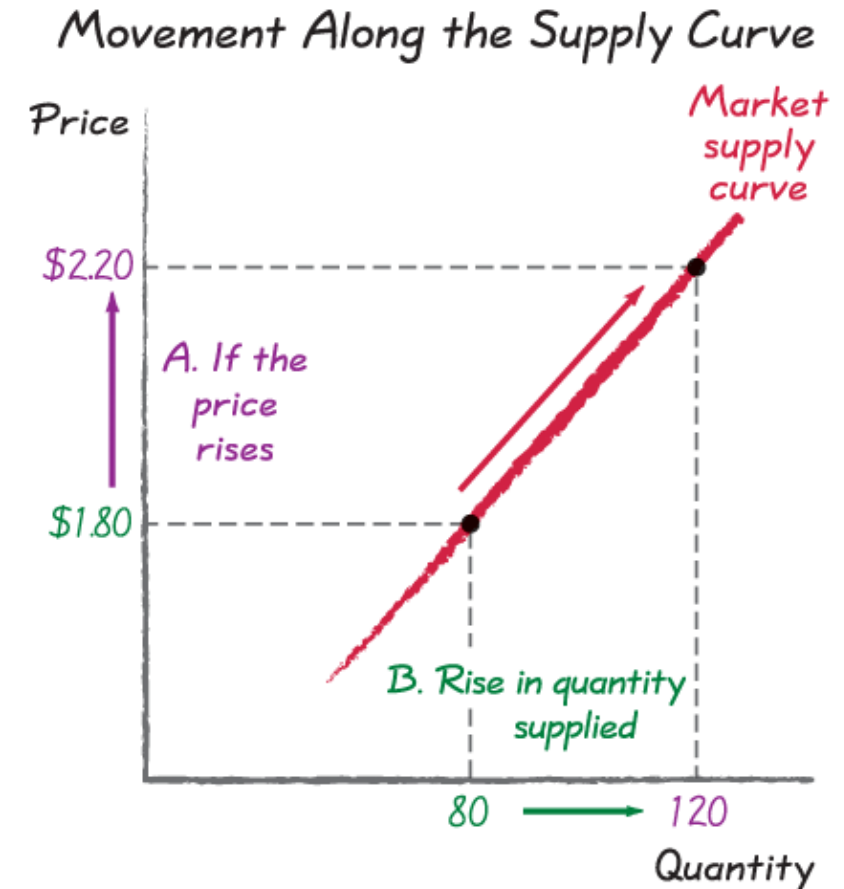


	A Individual supply	×	B Number of suppliers	=	C Market supply
Price (per litre)	Quantity supplied per individual business (millions of litres per day)		Number of sellers in the market		Total quantity supplied (millions of litres per day)
\$2.40	14	×	10	=	140
\$2.20	12	×	10	=	120
\$2.00	10	×	10	=	100
\$1.80	8	×	10	=	80
\$1.60	6	×	10	=	60



Movements along the supply curve

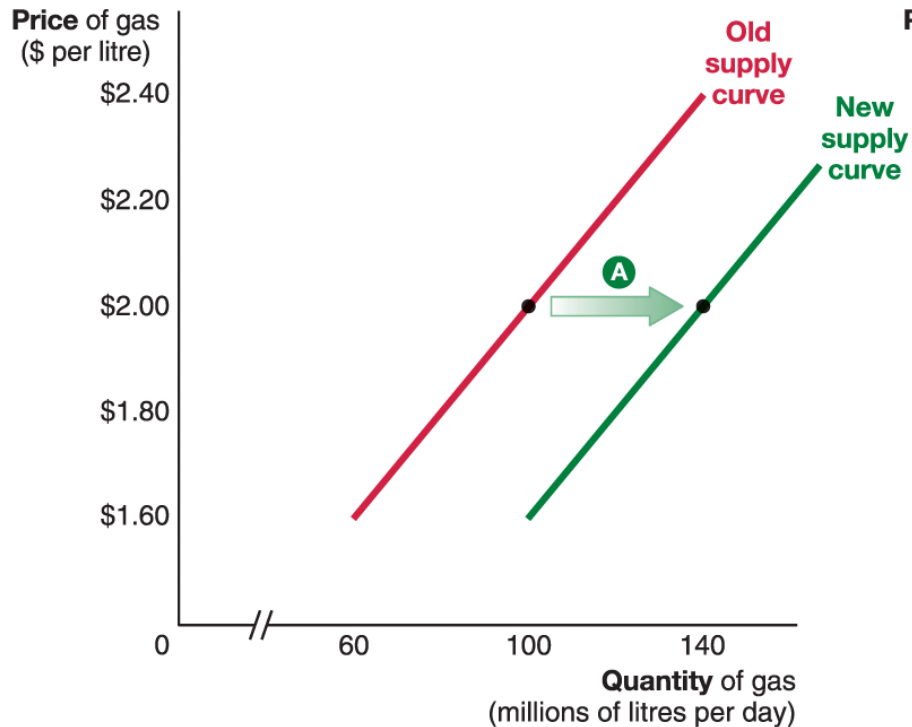
- A change in the own price causes **movement along the supply curve** yielding a **change in the quantity supplied**.
- A higher price leads existing firms to expand their production and new firms to enter the market. Both factors serve to increase the quantity supplied in the market.



Shifts in the market supply curve

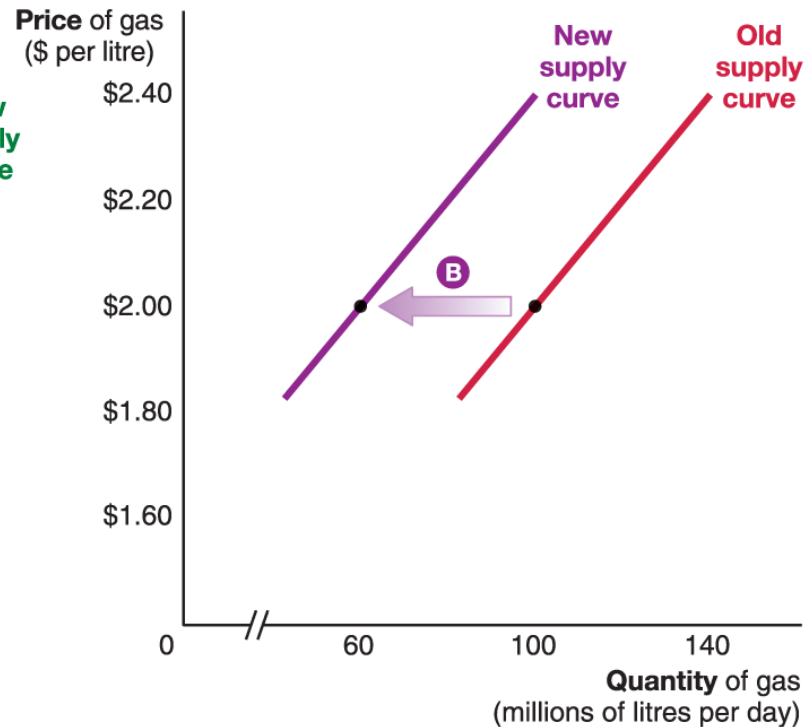
Panel A: An Increase in Supply

- A** An **increase in supply** shifts the supply curve to the right, leading to a higher quantity supplied at each and every price.



Panel B: A Decrease in Supply

- B** A **decrease in supply** shifts the supply curve to the left, leading to a lower quantity supplied at each and every price.



What happens to the **market supply curve** of gasoline if the market price of crude oil increases?

Five factors that shift the supply curve

1. Input prices

- An increase in the price of the inputs used to produce output increases the marginal cost of production and reduces the number of units supplied at every price.

2. Productivity and technology

- Productivity gains occur when businesses learn how to produce more using the same inputs. May reflect technological change (new machinery or equipment), new management techniques or production processes, or learning by doing.

3. Prices of related outputs

- **Substitutes-in-production** are alternative outputs produced using the same inputs, whereas **complements-in-production** are produced together.
- An increase in the price of a substitute-in-production causes a supply to decrease, whereas an increase in the prices of a complement-in-production causes supply to increase.

Five factors that shift the supply curve

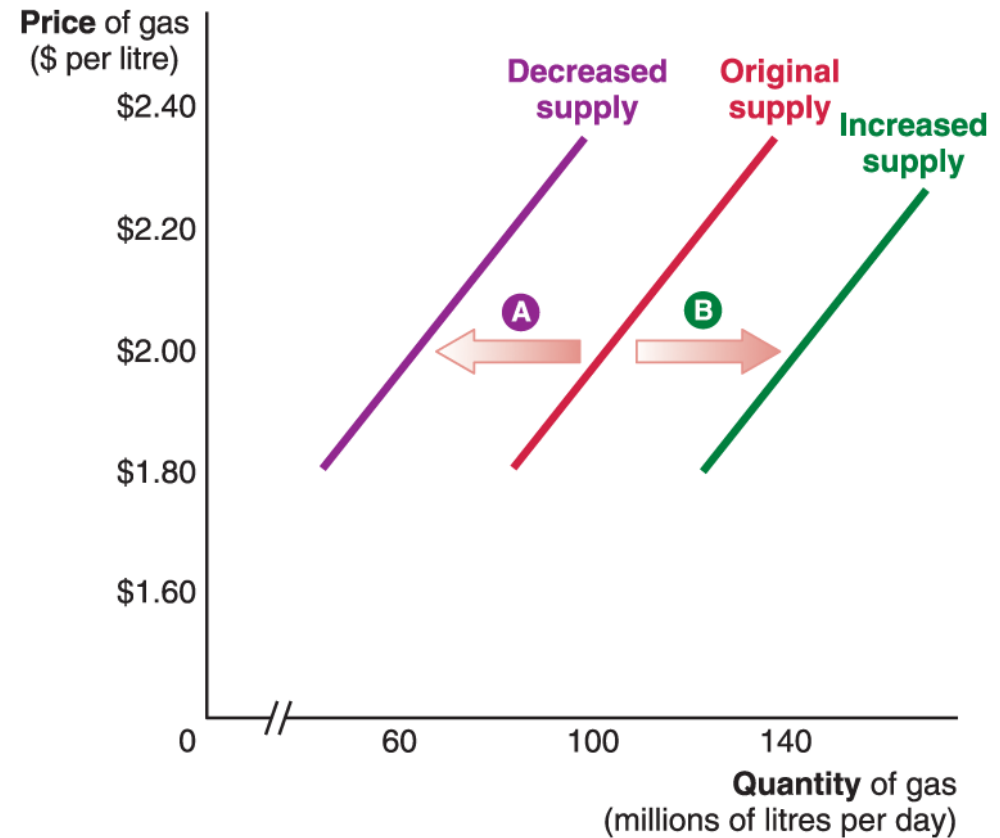
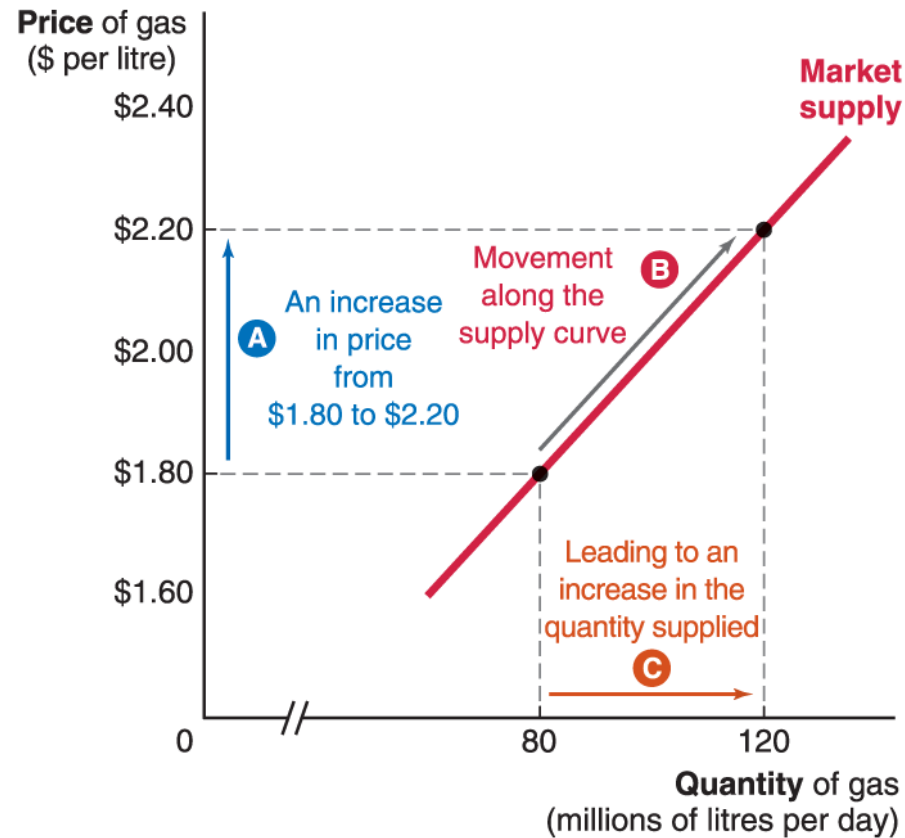
4. Expectations

- A belief that the price of the output will increase in the future incentivizes firms to store the output instead of selling it, thereby reducing supply.
- More relevant for outputs that are non-perishable (gasoline versus fresh fish).

5. The type and number of sellers

- A change in the number of sellers in the market affects market supply.
- A ruling by the Competition Bureau to allow or disallow a corporate merger affects the number of sellers in the market and potentially supply and, in turn, the market price.
- Unlike the first four factors, this factor shifts the market supply curve, but not individual supply curves.

Movements along vs shifts of the supply curve



Instructions for Term Test #1

1. Go to the toilet in advance. You won't be permitted to use the washroom during the test.
2. Bring an HB pencil, eraser, basic calculator, and your student ID card.
3. Arrive at your assigned classroom at **4:30pm** on **Friday October 2**. Wait outside the classroom until someone comes to let you in.
4. Promptly find a seat with a question booklet and answer sheet. Fill in your name, test version (A or B), and student number on the answer sheet.
5. Wait for the announcement to begin writing the test.
6. There are 25 multiple-choice questions on the test. You will have 80 minutes to complete the test.
7. For some questions, more than one answer may have an element of truth. In these cases, you should select the alternative that provides the “best” answer.
8. No clarification questions will be answered during the test.

Instructions for Term Test #1 (continued)

9. Scrap paper is not permitted. Do all your calculations and reasoning on the question booklet.
10. You are allowed to leave when you are finished. Hand in your answer sheet and question booklet separately. Leave quietly.

Last name begins with:	Assigned classroom:
A	STC 0010
B-C	MC 1085
D-F	STC 0020
G-H	STC 0040
I-M	EXP 1689
N-V	STC 1012
W-Z	STC 0050